FlashWatch: Design Spec

Oct 17, 2026 · @Jamie

Summary

FlashWatch forecasts the onset of flash drought 1 to 4 weeks ahead on a 25 km grid over the US Corn Belt and Plains. It turns those forecasts into county advisories using a Meta model agent that checks its own track record.

The claim to prove in 10 days: **calibrated onset probabilities that beat climatology, persistence and raw ECMWF forecasts on held-out years, at least at 1 to 2 week leads.** We also measure honestly whether skill holds at 3 to 4 weeks.

The design takes two lessons from the conflict model:

- **Forecast onset, not continuation.** Cells already in drought are excluded from the target, so persistence can't inflate skill.
- **Keep probabilities calibrated.** No downsampling, a plain log-loss objective, isotonic calibration per lead, and reliability diagrams in every report.

Deliverables by Nov 4:

1. An ingestion and feature pipeline with strict as-of dates and data latency built in.
2. A cross-validated skill report: Brier skill, reliability and an ablation of land state against forecast inputs.
3. A weekly agent that reads maps and tables, writes advisories and scores its own past forecasts.
4. A replay demo of the 2023 Midwest flash drought, week by week, alongside the US Drought Monitor.

Scope and success criteria

The target is a yes/no event per grid cell: does flash drought onset begin in a given week after the forecast issue date? Each lead is scored separately.

Item	Choice	Why
Region	36–49°N, 104–80°W (Corn Belt and Plains), about 5,000 cells at 0.25°	Covers the 2012, 2017 and 2023 events; cropland gives clear impacts

Item	Choice	Why
Years	2004–2025 growing seasons	Matches the 20-year ECMWF reforecast window, so every model sees the same years
Season	Issue dates Apr 1 – Oct 15	Flash drought matters for crops and is identifiable in warm-season soil moisture
Issue dates	Mondays and Thursdays	Matches ECMWF reforecast start days; about 55 per season
Leads	Week 1, 2, 3, 4 after issue, plus "any onset in days 1–28"	Week-by-week skill shows where it fades; the 28-day window is the headline number
Eligible cells	Root-zone soil moisture above the 20th percentile at issue	Removes cells already in drought (onset only)

Primary metric: Brier skill score (BSS) against climatology, out of fold, on eligible cells. Confidence intervals come from a bootstrap that resamples whole years.

Success:

- BSS above 0 at weeks 1–2, with the 95% CI excluding 0.
- Reliability diagrams within the bootstrap band at each lead.
- The land-state vs. forecast-input ablation reported with CIs at weeks 3–4, whichever way it comes out.
- In the 2023 replay, the week-2 probability crosses the alert threshold before the Drought Monitor's first one-category worsening over the core area.

Stretch: BSS above 0 at weeks 3–4. Published work suggests this is hard, so a clear null is a legitimate result.

Data sources

ERA5-Land is the training backbone, chosen for one consistent land-state record across all 22 years. SMAP is the JPL experiment: does an observation-constrained land state beat the reanalysis on the same years? ECMWF reforecasts supply the forecast inputs.

Dataset	Access route	Resolution and period	Role	Account	Caveats
ERA5-Land (soil water layers 1–4, 2 m temperature and dewpoint, precipitation, evaporation, potential evaporation, radiation)	Earth Engine `ECMWF/ERA5_LAND/DAILY_AGGR`; or the CDS ARCO data lake (geo-chunked Zarr, beta since Jun 2026); or CDS requests	0.1°, hourly/daily, 1950–present	Land-state features; onset labels	Earth Engine (noncommercial) or CDS	About 5 days latency; a model, not observations
SMAP L4 soil moisture `SPL4SMGP.008` (surface and root zone)	Earth Engine `NASA/SMAP/SPL4SMGP/008`; or AppEEARS area request; or NSIDC with Harmony subsetting	9 km, 3-hourly, Apr 2015–present	SMAP-era input variant (2015–2025); label cross-check	Earthdata (and Earth Engine if used)	Global granules are large, so subset on the server; short climatology for percentiles
ECMWF extended-range reforecasts (S2S)	ECMWF Data Store, dataset `s2s-reforecasts`, via the CDS client with the CDS key	1.5°, daily to 46 days; 11-member reforecasts covering about 20 years before each model cycle	Forecast features (precipitation, temperature, dewpoint, soil moisture, evaporation)	ECMWF (free)	Use one model cycle's reforecasts throughout; real-time S2S archive is delayed
NOAA GEFSv12 reforecast	AWS Open Data	0.25–0.5°, 2000–2019, to 35 days	Fallback forecast source	None	Different period and ensemble; large files
US Drought Monitor	USDM data services / weekly shapefiles	Weekly, 2000–present	Real-world check on labels; demo comparison	None	Expert blend, designed to be conservative
SMVI flash drought inventory	HydroShare download	Global 1990–2021 (CONUS 1979–2018 version)	External label check	Possibly HydroShare	Ends 2021; based on reanalysis
USDA NASS Crop Progress and Condition	Quick Stats API	Weekly, state level	Agent cross-check; impact evidence	Free API key	State level only
ECOSTRESS V2 evapotranspiration and ESI	AppEEARS	70 m, irregular overpasses, 2018–present	Demo visuals for 2023; spot checks	Earthdata	Patchy coverage; V1 retired

Dataset	Access route	Resolution and period	Role	Account	Caveats
Static layers: soil water capacity (gNATSGO/SoilGrids), cropland (USDA CDL), county boundaries (Census TIGER), Niño 3.4 (NOAA CPC)	Direct downloads	Static / monthly	Features; aggregating to counties	None	Use the CDL year before the issue date

Recommended route: Earth Engine for ERA5-Land and SMAP. It subsets and aggregates on the server, which avoids downloading terabytes of global SMAP granules. Each source has a backup route in case Earth Engine access stalls.

Access test, Oct 17 (from your PC): every source is reachable with working credentials. The only failure is the Drought Monitor statistics API, which we don't need.

Source	Result	What it means
Earth Engine	Pass: ERA5-Land daily has all 150 bands (none missing); 8 SMAP v008 images on the test day	Primary route for ERA5-Land and SMAP confirmed
ECMWF Data Store (S2S reforecasts)	Pass: 1 member, 1 variable, 46 days in 23 KB; 191 s, mostly queue	The old Web API closed in May 2026, so use the Data Store's `s2s-reforecasts` with the CDS key. Bundle members, variables and hindcast years per request
Copernicus CDS (ERA5-Land)	Pass: one day in 133 KB; 86 s, mostly queue	Too slow for 22 years of requests; backup only
SMAP L4 via NASA	Pass: 153 MB granule in 33 s; 8 granules a day	A full download would be about 4.6 TB, so subset on the server
NASA Earthdata and AppEEARS	Pass	ECOSTRESS and SMAP backup routes available
Drought Monitor weekly shapefiles	Pass: 1.7 MB per week	About 2 GB for 2004–2025
Drought Monitor statistics API	Empty response	Not needed
SMVI inventory (HydroShare)	Pass: 21 files, 5.8 GB in 5-year archives	Download only 2004–2021

Source	Result	What it means
USDA Quick Stats	Pass: 115 rows for Illinois corn condition, 2023	Agent cross-check data available
NOAA GEFSv12 reforecast (AWS)	Pass	Fallback forecast source is reachable
Muse Glimmer (Hugging Face)	Pass: full weights and GGUF builds	Runs locally in Ollama or LM Studio

Reforecast coverage to resolve: reforecasts issued with the 2024 forecasts cover roughly 2004–2023. Forecast inputs for the 2024–2025 test years must come from the real-time archive, subsampled to 11 members to match, or from the next model cycle's reforecasts. Decide on Day 1.

Common grid and data cube

Everything lands on one 0.25° latitude-longitude grid (52 × 96 cells) with daily time steps. The whole training cube fits in a few GB, so the full pipeline runs on a laptop.

- **Why 0.25° rather than 9 km:** soil moisture anomalies vary smoothly at this scale, Midwest counties are 30–40 km across, and one lead's training table drops from about 20M rows to about 3M.
- **Regridding:** area-weighted means from finer sources (ERA5-Land 0.1°, SMAP 9 km). Forecasts at 1.5° are interpolated to the grid for convenience, recorded as adding no information. Static layers use area-weighted means, or cropland fraction for CDL.
- **Time:** daily means (daily totals for precipitation and evaporation), dated in UTC. Labels use pentads (5-day means) to damp daily noise. Forecast features are aggregated to week 1–4 means relative to the start date.
- **Storage:** one Zarr store for observed fields (`time, lat, lon`), one for forecasts (`init, week, stat, lat, lon`), where `stat` is ensemble mean, spread and P(below tercile). Training tables are Parquet files, one per lead.

Store	Contents	Estimated size
`obs.zarr`	ERA5-Land, about 10 variables, 2004–2025, daily	~1.6 GB
`smap.zarr`	Surface and root-zone soil moisture, 2015–2025, daily	~0.2 GB

Store	Contents	Estimated size
`fcst.zarr`	Reforecasts, ~8 variables × 4 weeks × 3 statistics, about 2 starts a week	~0.5 GB
`train_lead{1..4}.parquet`	Cell × issue date rows with features, labels and fold IDs	~1 GB each

Size estimates are back-of-envelope; the day-1 access test replaces them with real numbers.

Labels

Onset is a rapid fall in root-zone soil moisture percentile: from at least the 40th to at most the 20th within 20 days, then staying low for 10 days. This is a daily-resolution version of the widely used percentile-drop definition. Two independent sources check it.

Primary definition. Let P(d) be the percentile of the 5-day running mean of root-zone soil moisture (0–100 cm) on day d. It is computed against the same calendar window (±15 days) in other years. Onset day d satisfies all three conditions:

1. P(d) ⩽ 20, and P was above 20 the day before.
2. P ⩾ 40 on at least one day in [d − 20, d − 5].
3. P stays ⩽ 20 for all of [d + 1, d + 10]. Labels may use future data; features never do.

Masks. Exclude cell-days with snow cover or frozen soil (ERA5-Land snow depth and soil temperature), and cells under 20% vegetated land.

Percentiles without leakage. Climatologies are rebuilt for each cross-validation fold from the training years only. With 21 years per fold, each day-of-year window holds about 600 values.

Lining labels up with forecasts. For issue date τ and lead week w, the label is 1 if an onset day falls in (τ + 7(w−1), τ + 7w]. The 28-day label is the OR of weeks 1–4. Rows exist only for eligible cells, those with P(τ − latency) > 20.

Episodes. Onset days are grouped into space-time clusters (8-connected cells, within ±7 days). Episodes are the unit for event-level scoring and for stating how many independent events the data really contains.

Cross-checks (Day 2):

Check	Source	Pass condition
Known events appear	2012 central US, 2017 Northern Plains, 2022, 2023 Midwest	Each is a large episode near the documented date
Agreement with a published inventory	SMVI inventory, 2004–2021	Episode-level overlap well above chance; disagreements explained
Real-world relevance	Drought Monitor worsening of 2+ categories within 4 weeks	Most large episodes are followed by worsening
Independent soil moisture	SMAP L4, 2015–2025	Onset days within ±7 days for most episodes
Sensitivity	Thresholds 30/20 and 40/10	Base rate and episode list stable enough that conclusions hold

The base rate per eligible cell-week is unknown until Day 2. My rough guess is 1–3%, which is low enough that calibration needs care.

Features

Every feature is computed as of the issue date minus that source's real-world delay. A hindcast should never see data a forecaster couldn't have had that morning. One function, `as_of(source, issue_date)`, enforces this, and a unit test fails if any feature reads past it.

Group	Features (about 60 in total)	Source	Delay applied
Soil moisture state	Root-zone and surface percentiles; change over the last 5, 10 and 20 days; days since the last time above the 40th percentile	ERA5-Land (or SMAP in the variant)	5 days (SMAP: 3)

Group	Features (about 60 in total)	Source	Delay applied
Atmospheric demand	Anomalies over the last 14 and 30 days of temperature, vapour pressure deficit, potential evaporation and radiation	ERA5-Land	5 days
Water balance	Precipitation deficit over 14, 30 and 60 days (SPI-style); evaporation divided by potential evaporation (an ESI proxy)	ERA5-Land	5 days
Forecast, per week 1–4	Ensemble mean and spread of precipitation, temperature, vapour pressure deficit and soil-moisture anomalies; P(precipitation in lower tercile)	ECMWF reforecast starting on the issue date	0 days
Season and climate mode	Day of year (sine and cosine); Niño 3.4	Calendar; NOAA CPC	Latest monthly value published before the issue date
Static	Soil water capacity; cropland fraction; aridity index; 30-year mean growing-season precipitation	gNATSGO/SoilGrids; CDL; ERA5-Land climatology	CDL from the year before

Rules:

- No latitude, longitude or cell ID as features, so the model can't memorise places. Static climate descriptors carry the geography.
- Anomalies and percentiles use each fold's training-year climatology, the same as the labels.

- Forecast anomalies are taken against the reforecast's own climatology for each start date and week. This removes model drift and bias.
- Missing data stays missing (LightGBM handles NaN natively). A missing-rate report runs per feature and year.

Models

Models form a ladder in which each rung must beat the one below on out-of-fold Brier score. The main model is gradient-boosted trees, with one model per lead. A deep spatial model is only built if the trees leave obvious skill on the table.

Rung	Model	Inputs	Purpose
B0	Climatology: smoothed onset rate per cell and week of year	None	Reference for Brier skill
B1	Logistic regression on current percentile and 10/20-day trend	Soil moisture state only	Persistence-plus-trend benchmark
B2	Raw dynamical: apply the onset rule to the reforecast's own soil-moisture forecast, then fit a logistic recalibration	Reforecast soil moisture	Answers "why not just use ECMWF?"
M1	LightGBM	Land state, water balance, season, static	Best land-only model
M2	LightGBM	M1 inputs plus forecast features	The main model
M3 (stretch)	Small U-Net or ConvLSTM on 32 × 32-cell patches	Same fields as M2, as images	Only if M2 shows spatial structure in its errors

Training choices:

- Plain binary log loss with no class weights or downsampling, so outputs stay close to probabilities. This avoids the conflict-model problem of predicted rates running at 0.37 of observed.
- Monotone constraints where physics is clear: lower current percentile and larger precipitation deficit never decrease risk.

- Hyperparameters tuned inside the training folds only (nested), with a small search of about 30 trials per lead.

Calibration: isotonic regression per lead, fit on out-of-fold predictions from the inner folds, then applied to the outer fold. Platt scaling is kept as a fallback if isotonic overfits the rare-event tail.

The ablation that matters: M2 minus M1, per lead, with year-bootstrap CIs. Published ML work found subseasonal skill comes mostly from the current land state (Lorenz & Otkin). Newer hybrid models report gains from adding dynamical forecasts beyond two weeks (Lesinger & Tian 2025). Testing this on onset specifically is a real contribution either way.

Explanations: SHAP values per prediction, aggregated to the top three drivers per county. These feed the agent's advisories ("risk is driven by a 30-day rainfall deficit and a forecast warm week 2").

Evaluation

Whole years are held out, because every cell in a season shares the same weather. 2004–2022 is used for leave-one-year-out cross-validation, and 2023–2025 is a test set opened once, after the model is frozen on Day 5.

Splits:

- **Development:** 19 folds, each holding out one season (2004–2022). Tuning and calibration are nested inside the training years.
- **Test:** 2023, 2024 and 2025, scored once with the frozen M2 and all baselines. The 2023 demo replay comes from this set, so it's genuinely out of sample.
- **SMAP variant:** 2015–2022 cross-validation, compared with M1/M2 on exactly the same years and cells.

Metrics, per lead and for the 28-day window:

Metric	What it answers
Brier score and Brier skill score vs. B0	Overall probabilistic accuracy (primary)
Reliability diagram plus calibration slope and intercept	Do 20% forecasts verify 20% of the time?
Area under precision-recall curve, alongside base rate	Ranking skill for a rare event (ROC AUC flatters rare events)
Hit rate and false alarm ratio at a fixed alert threshold	Operational trade-off for a user

Metric	What it answers
Episode lead time	For each episode, days between the first alert over most of its area and the onset
Drought Monitor lead	Days between the alert and the Monitor's first one-category worsening over the same cells

Uncertainty: 1,000 bootstrap resamples of whole years for every headline number. Paired bootstraps for model differences (M2 − M1, M2 − B2).

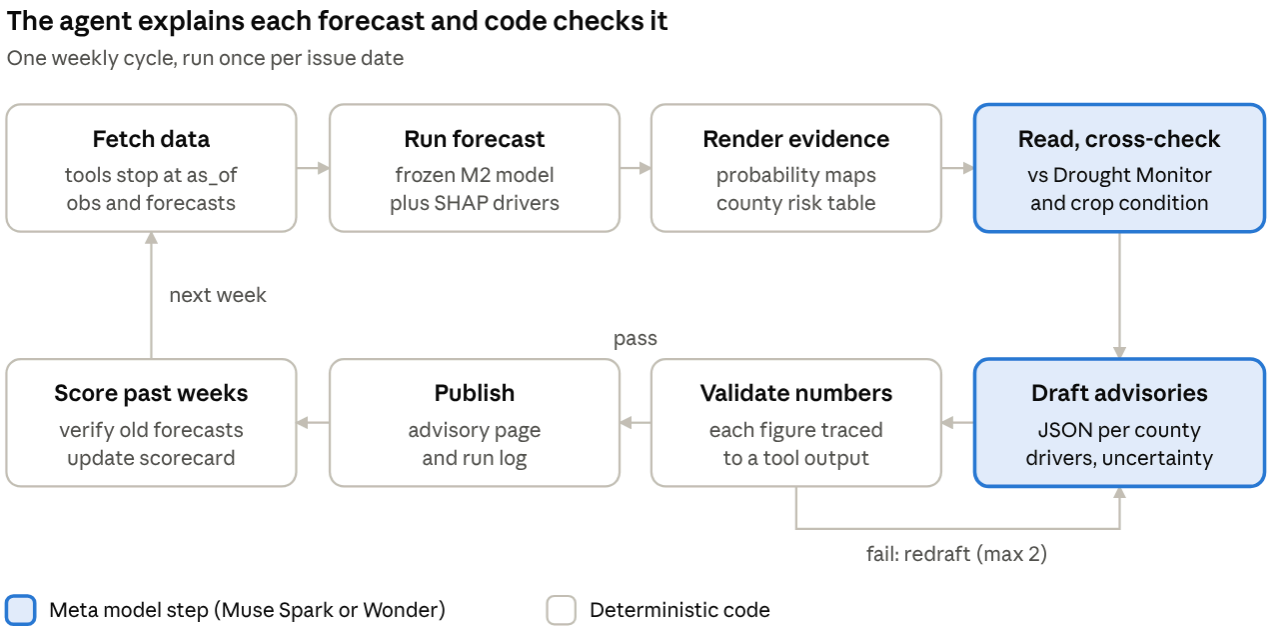
Report contents: a skill-by-lead table with CIs, reliability diagrams per lead, maps of skill across the region, the ablation, case studies (2012 out of fold, 2023 test), and the number of independent episodes behind each figure.

Pre-registered expectations, written before the first run so results can't be rationalised afterwards:

1. M1 beats B1 at weeks 1–2.
2. M2 beats M1 at weeks 3–4 but not at week 1.
3. B2 is poorly calibrated before recalibration.
4. Skill is higher in the western Plains than in the humid east.

Agent layer

The agent explains and checks; it never forecasts. Every probability comes from the frozen model. The agent turns those numbers into advisories an extension agent or water manager can act on, and code verifies every figure it writes.



weekly agent cycle · 8 steps, 2 by the Meta model

Only two of the eight steps use the Meta model. Everything else is deterministic, so a bad generation can't change a forecast.

Tools (function calls, all taking an `as_of` date and refusing later data):

Tool	Returns
`get_forecast(as_of)`	Probability per cell and lead; county averages and maxima
`get_drivers(as_of, county)`	Top three SHAP drivers with their values ("30-day rainfall at 18th percentile")
`get_map(as_of, layer)`	PNG of forecast probability, soil-moisture percentile or Drought Monitor categories
`get_drought_monitor(as_of)`	Latest county categories and that week's regional narrative text
`get_crop_condition(as_of, state)`	Percent of corn and soybeans rated poor or very poor, and the weekly change
`get_scorecard(as_of)`	Verified forecasts issued 28+ days earlier: hits, misses, false alarms, Brier to date
`publish_advisory(json)`	Runs the validator, then writes the advisory

Advisory schema, per county: FIPS and name; risk level (Low, Elevated, High, Severe, from fixed probability bands); probability for weeks 1–4; top three drivers; a one-sentence confidence statement; what to watch next week; actions for two audiences (producers and extension; water managers); the tool calls it relied on.

Number validator: pulls every number out of the advisory text. Each must match a tool output within rounding, or the draft is rejected with the offending figure named. After two failed redrafts, the advisory publishes with the flagged sentences removed.

Self-scoring: each cycle the scorecard checks forecasts from 28+ days earlier against observed onsets. The weekly summary must cite it, misses included.

Model choice: Muse Spark 1.1 or Muse Wonder through the Meta Model API, for image input, parallel tool calls and structured outputs. Muse Glimmer runs locally through the same client during development. Rough cost: one cycle is about 50–100k input tokens including images, so a full 18-week replay costs a few dollars on Spark and well under the $150 credit on Wonder. This is an estimate, to be replaced once prompts exist.

Stretch, a narrow fine-tune: LoRA on Muse Glimmer (Modal credits). It learns to write Drought Monitor-style regional narratives from each week's structured state, trained on about 1,000 weekly pairs from 2004–2022 and judged blind against real 2023–2025 narratives. First check that the narrative archive can be collected with light, polite downloading.

Demo

The demo replays the 2023 Midwest flash drought one issue date at a time, from May 1 to Aug 31. At each date the viewer sees only what FlashWatch knew that morning, next to what actually happened.

What the viewer sees, per issue date:

- **Left panel:** FlashWatch week-2 onset probability map, with the alert threshold outlined.
- **Right panel:** onsets that actually happened in the following two weeks, with the Drought Monitor's category changes overlaid.
- **Below:** the agent's advisories for the counties at highest risk, exactly as written at that date, plus its running scorecard of past forecasts.
- **Timeline strip:** for a few focus counties, the dates FlashWatch alerted against the dates the Monitor worsened.
- **One close-up:** an ECOSTRESS 70 m evapotranspiration or stress image over fields in the core area, if overpass coverage allows.

Keeping the replay honest:

- Every agent tool takes an `as_of` date and refuses data after it.
- The LLM may already know the 2023 drought from its training data. Replay mode passes relative dates ("week 22") and anonymised county IDs, and prompts forbid outside knowledge. A validator rejects any advisory number that isn't in a tool output.
- The model behind the replay is the frozen Day-5 M2, scored once on the test years.

Second case: 2012 central US, from out-of-fold cross-validation predictions, as a shorter clip.

Submission package:

1. Public GitHub repository with README, results table and one-command reproduction of the report from the cached cube.
2. A 3-minute video: problem (20 s), 2023 replay (90 s), skill and reliability (40 s), agent close-up (30 s).
3. The replay as a static web page.
4. A two-page write-up and a model card: data, labels, skill, limitations, intended use.

Video length and format follow the official rules once published.

Repository and pipeline

One Python package with a command-line entry point per stage. Each stage reads and writes files on disk, so any step can be rerun alone or moved to another machine. Everything except the optional fine-tune runs on a laptop CPU.

```
flashwatch/
  config/        region.yaml, sources.yaml, labels.yaml, model.yaml
  ingest/        era5land.py, smap.py, s2s.py, usdm.py, smvi.py, nass.py,
static.py
  grid/          regrid.py            # everything to the 0.25° grid
  labels/        onset.py, episodes.py
  features/      asof.py, build.py    # as_of() and latency table
  models/        baselines.py, gbm.py, calibrate.py, explain.py
  eval/          cv.py, metrics.py, bootstrap.py, report.py
  agent/         llm.py, tools.py, loop.py, schema.py, validate.py, prompts/
  demo/          replay.py, site/
  scripts/       check_access.py
  tests/         test_asof.py, test_labels.py, test_metrics.py
```

Command	Does	Runs on
`fw check-access`	Fetches one small sample from each source and reports status, size and coverage	Laptop (now)
`fw ingest <source> --years 2004-2025`	Downloads and regrids into the Zarr stores	Laptop or Modal
`fw labels`	Builds onset days, episodes and cross-check report	Laptop
`fw features --lead 1..4`	Builds Parquet training tables with fold IDs	Laptop
`fw train --model m2 --cv`	Nested CV, calibration, out-of-fold predictions	Laptop (LightGBM, minutes per lead)
`fw report`	Skill tables, reliability, maps, bootstrap CIs	Laptop
`fw agent --as-of 2023-06-12`	Runs one weekly agent cycle	Laptop plus Meta Model API
`fw replay --season 2023`	Runs the agent across all issue dates and builds the demo site	Laptop plus Meta Model API

LLM client: `agent/llm.py` wraps any OpenAI-compatible endpoint (`base_url` , `model`). Development can use Muse Glimmer locally until Meta Model API access arrives. Switching is a config change.

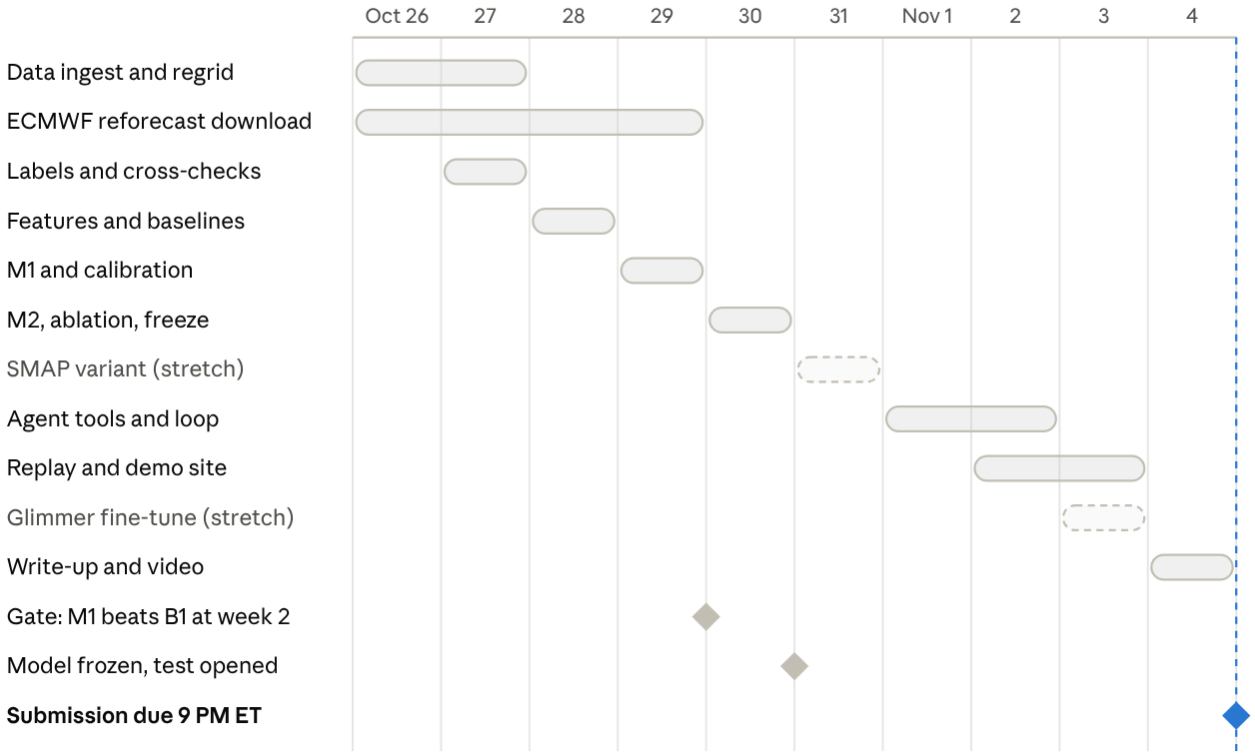
Execution: link this chat to the desktop app on your computer and I can run these commands there. Modal is the alternative for long downloads.

10-day schedule

The model is frozen at the end of Day 5 (Oct 30), so the second half goes to the agent and the demo, where judges spend their attention. Stretch items are cut first: the fine-tune, then the SMAP variant.

The model is frozen on day 5, leaving five days for the agent and demo

Hackathon days, US Eastern dates, 2026. Dashed bars are cut first.



10-day schedule · Oct 26 – Nov 4, three gates

Exit criterion for each day:

1. **Oct 26:** `obs.zarr` built for the region and all years; reforecast requests queued; Drought Monitor and SMVI downloaded.
2. **Oct 27:** onset labels and episodes built; cross-check table filled; base rate known.
3. **Oct 28:** training tables for all four leads; `as_of` leakage test passing; B0–B1 scored in the CV harness.
4. **Oct 29:** M1 calibrated and scored. **Gate:** if M1 doesn't beat B1 at week 2, narrow the headline claim to week 1 and the 28-day window.
5. **Oct 30:** M2 and B2 scored; ablation with CIs; model frozen; test years 2023–2025 scored once.
6. **Oct 31:** SMAP variant (stretch); SHAP drivers aggregated per county; map rendering done.
7. **Nov 1:** agent tools, schema, validator and one full cycle on a single issue date.
8. **Nov 2:** cross-checks and self-scoring working; replay engine running the 2023 season.
9. **Nov 3:** demo site and both case studies; draft write-up; Glimmer fine-tune if time allows.
10. **Nov 4:** video, README, model card; submit by 9 PM ET (1 PM Nov 5 in Sydney), with at least 3 hours of buffer.

Never cut: calibrated out-of-fold evaluation, the as-of leakage test, and the 2023 replay.

Risks and open questions

The biggest risks are slow data queues on Day 1 and weak skill at weeks 3–4. The first is handled by testing access before kickoff; the second by framing a null result honestly.

Risk	Likelihood	Mitigation
ECMWF reforecast downloads are slow (many small requests, queued)	High	Start them at kickoff, batched per start date; GEFSv12 on AWS as fallback; M1 doesn't need them
Earth Engine or CDS access stalls	Medium	Backup routes listed per source; access test before kickoff
Little or no skill at weeks 3–4	High	Pre-registered; reported with CIs; the demo still works on week 1–2 skill
Too few independent episodes for tight CIs	Medium	Report episode counts; pool across the region; year bootstrap
Labels sensitive to thresholds	Medium	Two threshold variants; triangulate with SMVI, the Drought Monitor and SMAP
Reforecasts (11 members) differ from live forecasts (101 members, newer model)	Certain	Features use ensemble mean, spread and terciles, which are less sensitive to size; noted in the model card
LLM already knows how 2023 turned out	Medium	Anonymised replay, number validator, and prompts forbidding outside knowledge
Meta Model API access is late or region-limited (preview launched US-only)	Medium	Develop on local Muse Glimmer through the same client; confirm the key works from Australia on acceptance
Official rules restrict pre-built work	Unknown until Oct 26	Only spec and access checks before kickoff; all project code written from Oct 26
Scope creep (deep model, fine-tune)	High	Cut lines in the schedule; M3 and fine-tune are stretch only

Open questions:

- [ ] Which ECMWF model cycle's reforecasts to use, and exactly which years they cover.
- [ ] Does Earth Engine's ERA5-Land daily collection include every needed variable (potential evaporation, soil temperature)?

- [ ] What does the Meta Model API cost per full replay (about 18 issue dates × one agent cycle)? Estimate once prompts exist.
- [ ] Do the official rules require a specific Meta model to be central, or is the agent layer enough?

Pre-kickoff checklist

Accounts take minutes to create but can take days to approve, so register them now. Everything below is preparation, not project code.

Key times in Sydney time:

Event	US Eastern	Sydney (AEDT)
Applications close	Oct 25, 11:59 PM EDT	Oct 26, 2:59 PM
Kickoff livestream	Oct 26, 12:00 PM EDT	Oct 27, 3:00 AM
Submissions due	Nov 4, 9:00 PM EST (after US clocks change Nov 1)	Nov 5, 1:00 PM

Accounts:

- [ ] Submit the hackathon application (earlier is better; acceptance is rolling).
- [ ] NASA Earthdata login, and authorise the NSIDC and LP DAAC applications in your profile.
- [ ] Google Earth Engine: register a Cloud project for noncommercial use.
- [ ] ECMWF account for the sub-seasonal reforecast archive. This corrects my earlier note: the real-time open data needs none, but the archive likely does.
- [ ] Copernicus CDS account, and accept the ERA5-Land licence (backup route).
- [ ] USDA Quick Stats API key.
- [ ] Hugging Face account; check the Muse Glimmer download terms.

Access test (needs your computer or Modal):

- [ ] Link this chat to the desktop app on your computer, so I can run commands there.
- [ ] Run `check_access`: one small sample per source, reporting status, size, variables and date coverage. This is a throwaway diagnostic, not project code.
- [ ] Confirm the Earth Engine ERA5-Land variable list and the SMAP v8 collection over the region.
- [ ] Request one ECMWF reforecast start date and time how long it takes, to size the full download.

Decisions and rules:

- [ ] Read the official rules when published: pre-work, required Meta model use, licence, video format.
- [ ] Sign off this spec, especially the region, years, onset definition and pre-registered expectations.
- [ ] Once accepted, confirm the Meta Model API key works from Australia.